Wholesaling Automation Architecture v2

Oct 25, 2026 · @Aaron

Decisions locked

Supersedes Claude_Code_Wholesaling_Build_Guide_v1 where they conflict. Everything below is a decision made on 2026-10-25 unless marked open.

Area	Decision
System of record	Google Sheets as operator view; PostgreSQL underneath for idempotency, versioning, workflow state
Workflow engine	Temporal (Temporal Cloud)
Hosting	Temporal Cloud + Railway (workers, Postgres, webhook, web form). Claude Code deploys via CLI
Model layer	Provider-agnostic. Default Claude Haiku for classify/extract, Sonnet for drafting, QA, comp audit. Grok or others swappable per agent
Seller channels	Email, SMS (draft-only until 10DLC + TCPA cleared), inbound web form. No calls, no direct mail, no answering service
Boots on the ground	None. Virtual wholesaling: offers subject to inspection, seller-submitted photos, buyer due diligence
Markets	Multi-market from day one, selected by M1 from investor demand. Cap 3 to 5 Active at launch
Deal band	$75k to $350k purchase price (ARV ~$120k to $500k), per-market override
Budget	No hard ceiling. Soft cap + alert in config
Data sources	None yet. Provider selection is a build step

Area	Decision
Entity / legal	None yet. LLC, address, bank, title company are pre-launch checklist items. Operate unlicensed with TX equitable-interest disclosure
Volume targets	2,000 seller contacts/wk, 40 offers/wk, 4 contracts/mo, 2 to 3 assignments/mo at ~$10k avg
Buyer list	Starting from zero. Investor funnel runs first; seller outreach in a market waits for qualified buy boxes
Operator time	Unlimited during L0; promotion to L1 is on quality metrics only
Timeline	Build sets pace. ~4 weeks to L0 investor sends, seller side after
Zapier parser	Optional path behind a feature flag, default off

System architecture

The system is a set of Temporal workflows running on cloud workers that call a model API; Claude Code builds and deploys it but never runs it.

Stack

- Python 3.12, FastAPI (webhook endpoint, web form, review queue API)
- PostgreSQL (authoritative state), Google Sheets (operator projection, one-way sync from Postgres)
- Temporal Cloud for all long-running sequences, schedules, signals, retries
- Microsoft Graph for Outlook mail; Twilio (or equivalent) for SMS once registered
- Model provider abstraction (`agents/llm.py`) with per-agent model routing in config
- Railway: API container, worker containers, managed Postgres; secrets in Railway env
- Alerting via email digest + optional Slack webhook

Core principles

1. Postgres is the only writer of truth. Sheets is read by you, written by the sync workflow. Agents never write to Sheets directly.
2. Every outbound action goes through the same code path at every autonomy level; the policy engine decides whether the last step is `send` or `queue_for_human`.

3. Every agent run is logged: prompt hash, model, tokens, cost, structured output, decision.
4. Idempotency keys on every inbound event (Graph `internetMessageId`, SMS SID, form submission id) and every outbound send.
5. No autonomous contract execution, earnest-money transfer, or binding negotiation at any level.

Event paths

- Investor inbound: Graph webhook or 15-min delta poll → dedupe → A4 classify → draft → A5 QA → policy → send or queue → audit
- Seller inbound: email / SMS / web form → dedupe → S4 intake → if intake complete → `offer.requested` → A7
- Buy box: A4 extract → Postgres upsert (versioned) → `buy_box.updated` → D1 search profile → D2 match
- Deal: A7 offer accepted → `contract.requested` → human signs → S6 disposition to matched buyers

Autonomy policy

Autonomy is a per-action policy stored in a `policies` table, not a pilot phase. The code path is identical at every level; only the final step changes.

Action	L0	L1	L2
Investor initial outreach	Draft, human sends	QA-approved auto-send, daily cap	Auto-send, weekly review
Investor follow-ups	Draft	Auto if initial was approved	Auto
Reply to interested investor	Draft	Auto if QA confidence ≥ threshold, no price/contract/legal content	Auto
Seller outreach (email)	Draft	Auto, daily cap	Auto
Seller outreach (SMS)	Off	Draft-only until 10DLC + consent cleared	Auto with consent record

Action	L0	L1	L2
Seller intake replies	Draft	Auto	Auto
Offer generation and send	Draft	Auto if all A7 gates pass	Auto
Counter acceptance	Human	Auto within negotiation room	Auto within room
Contract execution	Human, always	Human, always	Human, always
Earnest money	Human, always	Human, always	Human, always
Buyer disposition blast	Draft	Auto for approved deals	Auto
Opt-out suppression	Auto	Auto	Auto

Promotion criteria (L0 → L1), per action

- ≥50 human-approved drafts with ≤5% requiring edits
- Zero QA-blocked drafts sent in the window
- Zero compliance flags (prohibited claims, opt-out violations)

Demotion: any compliance incident or edit rate >15% over 7 days drops the action one level automatically and alerts you.

Hard floors: contract, earnest money, anything with a dollar commitment outside A7's formula, and any legal question stay human at every level.

Agent roster

16 agents. Each is a Temporal activity: fixed system prompt, tool definitions, JSON output schema, model routed by config. A5 QA and the policy engine are shared across both funnels.

Agent	Side	Input → output	Default model
M1 Market Selection	Shared	Market signals → ranked markets (Active / Watch / Excluded)	Haiku + code
A1 Investor Discovery	Investor	Deed records in a market → scored prospects	Haiku + code
A2 Contact Research	Investor	Prospects → professional contact channels with provenance	Haiku
A3 Outreach	Investor	Template + contact → initial + 3 follow-ups in one thread	Sonnet
A4 Reply & Buy Box	Investor	Inbound reply → classification, draft reply, extracted criteria	Sonnet
A5 QA	Shared	Any outbound draft + context → approve / block / rewrite	Sonnet
A6 Parser Formatter	Investor	Confirmed buy box → Zapier parser email (flag off by default)	Haiku
D1 Search Profile	Deal	Buy box event → versioned search profile	Code
D2 Property Match	Deal	Profile + sources → ranked candidates with evidence	Sonnet + code
S1 Seller List Builder	Seller	Market + distress criteria → seller leads	Code
S2 Skip Trace	Seller	Owner → email, phone, mailing address with provenance	Code + vendor
S3 Seller Outreach	Seller	Template + lead → email/SMS sequence	Sonnet
S4 Seller Intake	Seller	Inbound reply / form → motivation, condition, timeline, asking, photos	Sonnet
A7 Offer Engine	Seller	Complete intake → verified offer + full calc record	Sonnet + code
S5 Negotiation	Seller	Seller response → counter handling, cadence, expiry, re-engage	Sonnet

Agent	Side	Input → output	Default model
S6 Disposition	Deal	Signed contract → buyer blast to D2 matches, deposit, closing coordination	Sonnet

M1 market selection

Markets are chosen by investor demand, re-scored monthly, and promoted Watch → Active only once qualified buy boxes exist there. Seller outreach never runs ahead of buyer demand.

Signals per metro or county

Signal	Measure	Weight
Investor demand	Share of sales to entities, cash purchase rate, repeat buyers (≥3 in 12 mo), investor-to-investor resales	0.30
Spread	Median cash-sale price vs. retail for similar sqft/age; median flip profit if available	0.25
Distress supply	Pre-foreclosure, tax delinquent, probate, absentee, vacancy per 1,000 parcels	0.20
Liquidity	Days on market, months of inventory, sold-count trend	0.10
Price band fit	Share of stock inside deal band	0.10
Legal friction	State wholesaling rules scored 0 to 3 (config table)	−0.05
Data availability	Records, comps, skip trace obtainable	Gate

`market_score = 0.30 demand + 0.25 spread + 0.20 distress + 0.10 liquidity + 0.10 price_fit − 0.05 friction`, after the data gate. Weights live in config and are retuned from realized results.

Tiers

- **Active:** top N (cap 3 to 5 at launch). Both funnels run.

- **Watch:** investor discovery and outreach only; building the buyer list. Promoted to Active at ≥10 qualified buy boxes.
- **Excluded:** failed legal or data gate.

Feedback loop: once live, proxy signals are replaced by own results per market (buy-box count, reply rate, offers accepted, assignments closed). A market drops to Watch if 90-day results fall below threshold.

Deal band per market: default global $75k to $350k; M1 may set a market's band at median × 0.5 to × 1.5, clamped to the global range.

Investor funnel

Find investors buying in an Active or Watch market regardless of where they live, qualify them on purchase behavior, and capture their buy box through a four-touch Outlook thread.

A1 qualification criteria (deed records in market)

- Buyer is an entity (LLC, LP, trust, corp) or an individual with ≥2 purchases in 24 months
- Cash or hard-money purchase (no conventional mortgage recorded)
- Residential 1 to 4 unit (multifamily and land configurable later)
- Purchase price inside the market's deal band
- Last purchase ≤18 months ago

Investor type (drives messaging and buy-box defaults)

Type	Signal	Treatment
Flipper	Resale within 4 to 12 months at higher price	Priority
Buy-and-hold	Holds >12 months, often clustered in one ZIP	Priority
Wholesaler / competitor	Assignment or double-close patterns, high velocity	Deprioritize
iBuyer / institutional	Known entity list	Exclude
Agent-investor	License match	Flag, include

Score (0 to 100): volume 30, recency 25, cash rate 15, price-band fit 15, market concentration 10, type bonus ±5. Score ≥50 → prospect, A2 runs. Score ≥75 → priority tier, first in outreach queue.

Exclusions: institutional, government, nonprofits, lenders taking REO, builders buying lots, suppression list.

Entity resolution: collapse LLC variants and shared mailing addresses into one investor record; registered agent or mailing address feeds A2. Every source deed is kept as evidence.

A3 sequence: approved Houston template parameterized by market and first name; initial + 3 follow-ups at ~4-day intervals in the same Graph conversation; subject-line tests on the initial only; any substantive reply, opt-out, or pause cancels remaining sends.

A4 classification: interested, buy box supplied, question, not now, not interested, opt-out, ambiguous, unrelated. Buy boxes are versioned; unknowns tracked; corrections overwrite with history.

Seller funnel

Seller outreach starts in a market only after it is Active. Channels are email, SMS (gated), and inbound web form; no calls, mail, or walkthroughs.

S1 list criteria (per Active market, refreshed weekly)

- Absentee owner (mailing ≠ situs), tax delinquent, pre-foreclosure, probate, code violations, long-term hold (≥15 yrs), vacant, expired listing
- Residential 1 to 4 unit, inside deal band on assessed or estimated value
- Score by stacked signals; ≥2 signals → outreach queue

S2 skip trace: vendor API; keep provenance and confidence per contact; suppress DNC and prior opt-outs before any SMS.

S3 sequence: email initial + 3 follow-ups at 3/7/14 days; SMS only once 10DLC registered and consent basis documented. Every seller-facing message carries the TX equitable-interest disclosure, sender identity, physical address, and opt-out.

S4 intake (reply, SMS, or web form)

Field	Required for offer
Property address (parcel-verified)	Yes

Field	Required for offer
Motivation	Yes
Condition class (Light / Medium / Heavy / Gut) + disclosed issues	Yes
Timeline to sell	Yes
Asking price	No
Photos (web form upload link sent automatically)	Requested; offer proceeds without, confidence drops one level
Co-owners, occupancy, liens, HOA	Asked; unknowns flagged

Intake is conversational: S4 asks only for missing required fields, one or two per message, and marks the lead `intake_complete` when all required fields are present. That event triggers A7.

S5 negotiation: auto-accept counters within negotiation room (offer to MAO); anything above MAO → human. Reminders at days 2/5/10; expire at 14; re-engage at 60/90. Declines capture reason and requeue at 90 days.

S6 disposition (after human-signed contract): blast to D2-matched buyers by tier order, collect proof of funds and deposit terms, assignment agreement via DocuSign, title/escrow coordination, deadline alerts (option period, inspection, closing), deal-death handling.

A7 offer engine

Every rule is numeric so no judgment call is left in the loop; an offer auto-sends only when every verification pass and every gate passes, and the full calculation is logged with it.

Pipeline

1. **Property resolution** — address → parcel ID; beds/baths/sqft/year/lot from county and provider.
2. **Comp pull** — sold comps: same ZIP, ±20% sqft, ±15 yrs, ⩽180 days, ⩽0.75 mi (tier 1). Expand only if fewer than 3: tier 2 = 1.0 mi / 270 d, tier 3 = 1.5 mi / 365 d. Record tier used.
3. **ARV** — comp price/sqft, trim top and bottom 15%, weight by recency and distance, × subject sqft.

- High confidence: ⩾5 comps, tier 1, spread ⩽12%
- Medium: ⩾3 comps, tier ⩽2, spread ⩽20%
- Low: anything else → human

4. **Repairs** — condition class × per-sqft rate table (config, per market) + disclosed line items (roof, foundation, HVAC). Foundation in Houston always adds a fixed contingency.
5. **MAO** — `ARV × market_pct − repairs − assignment_fee − closing_costs`. `market_pct` default 0.70 per market; fee = target per deal; closing = fixed % table.
6. **Offer** — MAO minus negotiation room (default 7%), rounded down to $500.
7. **Sanity gates** (all required for auto-send): ARV confidence ⩾ Medium; offer ⩽ market hard cap; offer ⩾ 35% of ARV; offer within ±25% of asking or asking blank; no open flags (title/lien, HOA, multi-unit, non-residential, probate without authority, co-owner missing); seller not suppressed; no active offer on same parcel.
8. **Draft** — locked template: number, "subject to inspection and signed agreement," cash close in N days, no fee disclosure, no buyer-lined-up claims. A5 QA checks facts and prohibited claims.
9. **Send** — per policy level. Full calc record attached.
10. **Post-offer** — hand to S5.

Verification loops (run between steps 6 and 7; max iterations fixed so the workflow always terminates)

Pass	Check	On mismatch
V1 Dual-source facts	County vs provider on sqft, beds, baths, year, lot	>10% gap → pull third source; majority wins; none → human
V2 Dual-method ARV	Comp calc vs provider AVM	>10% gap → re-run comps at next tier, re-compare; still >10% → confidence −1. Max 3 passes
V3 Comp audit	Model flags non-arm's-length, new build, already-renovated flip, lot-value outlier	Remove flagged, recompute; loop until no new flags or 2 iterations
V4 Repair cross-check	Intake class vs year built, last sale, listing photos/remarks	Evidence worse than stated → bump class one level, log reason

Pass	Check	On mismatch
V5 Independent recompute	Separate function recalculates MAO and offer from logged inputs	Must match to the dollar; mismatch → block, alert (code bug)
V6 Pre-send drift	Re-pull comps immediately before send	Any new comp moving ARV >5% → recompute, re-run gates

Verification record (attached to every offer): sources used, comps kept and rejected with reasons, ARV from both methods, repair class and any adjustment, every gate result. At L1 you review these records weekly, not individual offers.

Temporal workflows and triggers

Every multi-day sequence is a durable workflow; every recurring job is a Temporal schedule; every interrupt is a signal. Nothing depends on cron or a laptop.

Workflows

Workflow	Started by	Sleeps	Signals that interrupt
InvestorSequence	A2 qualified contact	4 d between sends	reply, opt_out, pause, bounce
SellerSequence	S1/S2 queued lead	3/7/14 d	reply, opt_out, pause, intake_complete
OfferWorkflow	intake_complete	none (runs verification then hands off)	human_decision
NegotiationWorkflow	offer sent	2/5/10 d, expire 14, re-engage 60/90	counter, accept, decline, human_override
DispositionWorkflow	contract signed	per deadline	buyer_response, deposit_received, deal_dead

Workflow	Started by	Sleeps	Signals that interrupt
BuyBoxSearch	buy_box.created/updated	none	superseded (newer version)

Schedules

Schedule	Cadence	Guardrail
M1 market rescore	Monthly	Promotion/demotion requires threshold crossed twice
A1 investor discovery	Daily 06:00 local per market	No overlapping runs; rate limits per provider
S1 seller list refresh	Weekly	Dedupe against existing leads and suppression
Graph subscription renewal	Every 48 h	Alert if renewal fails twice
Graph delta poll	Every 15 min	Reconciles missed webhooks; idempotent
Sheets sync	Every 15 min	One-way Postgres → Sheets
Daily reconciliation	Daily	Compare Outlook, Postgres, Sheets; alert on mismatch, never silently overwrite
Policy promotion check	Daily	Applies promotion/demotion criteria
Budget check	Hourly	Soft cap hit → pause discovery and search, keep inbound handling
Watchdog	Every 10 min	Heartbeats stale → alert

Resilience: circuit breakers on model API and Graph; dead-letter queue with retry and alert after 3 failures; idempotency keys on every send; workflows replay safely after worker restart.

Data, integrations, compliance

No data provider, entity, or sending identity exists yet; each is a build step with selection criteria, not an assumption.

Data providers to select (criteria: coverage in Active markets, API access, cost per record, license permits automated use)

Need	Candidates to evaluate	Feeds
Deed / transaction records	County APIs, ATTOM, PropStream, BatchData	M1, A1, S1
Distress lists	PropStream, BatchLeads, county tax/probate	S1
Skip trace	BatchSkipTracing, Tracerfy, PropStream	S2
Comps and AVM	ATTOM, HouseCanary, RentCast, MLS access via a licensed partner	A7, D2
Listings for D2	Same as comps, plus off-market from A7 contracts	D2

Integrations

- Microsoft Graph: mail read, send, subscriptions; app registered in Entra ID with least privilege; tokens in secrets manager
- Google Sheets API: service account, one-way write from sync workflow
- SMS: Twilio or equivalent; A2P 10DLC brand and campaign registration before first send
- DocuSign: PSA and assignment templates, pre-filled from deal record
- Web form: FastAPI page on Railway, photo upload to object storage, submission id as idempotency key
- Zapier parser: behind `ZAPIER_PARSER_ENABLED` flag, default false

Deliverability: dedicated sending domain(s) separate from your consulting domain; 2 to 3 week warmup before volume; SPF/DKIM/DMARC; bounce and complaint handling feeds suppression; domain rotation at scale.

Compliance

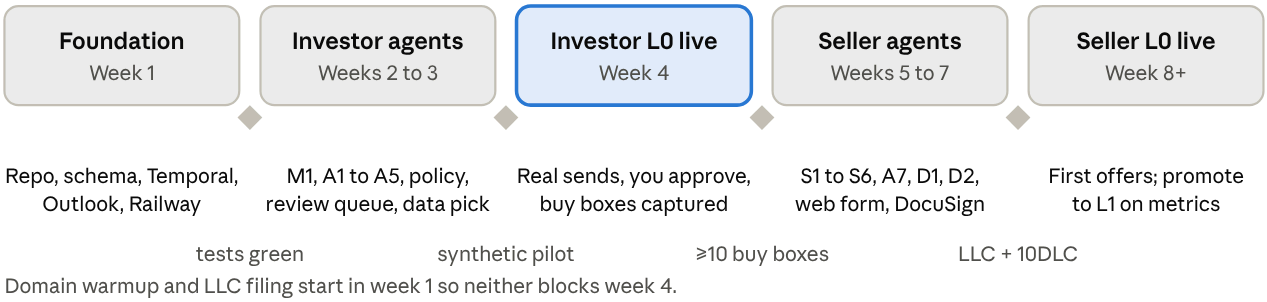
- CAN-SPAM: truthful headers, sender identity, physical address, working opt-out in every email

- TCPA / DNC: SMS off until 10DLC registered and consent basis documented per contact; DNC scrub before any SMS
- Texas: equitable-interest disclosure in all seller-facing material; no advertising a property you do not own or control; per-market legal review before activating a new state
- Privacy: minimize stored personal data, document sources, honor removal requests, no scraping of prohibited sources
- AI autonomy: no contracts, earnest money, binding negotiation, or property representations without human action

Build order

Investor side first so buyer demand exists before any seller outreach; each phase has a gate it must clear.

Investor sends go live in week 4; seller side follows once buy boxes exist



build roadmap · 5 phases, 4 gates

Phases are sequential; the two live phases run at L0 until promotion criteria are met.

Step detail

1. Repo, CI, `.env.example`, requirements matrix (REQUIREMENTS.md with CONFIRMED / PROPOSED / REJECTED / OPEN)
2. Postgres schema and migrations: markets, investors, contacts, buy_boxes (versioned), seller_leads, intakes, offers, offer_verifications, property_candidates, matches, messages, threads, sequences, policies, suppression, qa_reviews, agent_runs, audit_events
3. Temporal worker skeleton, one hello workflow, Railway deploy, health endpoint
4. Graph integration: app registration, webhook + delta poll, draft and send as separate operations

5. Model provider abstraction with per-agent routing and run logging
6. M1, A1, A2 with a mock data provider; real provider swapped in once selected
7. A3 InvestorSequence workflow; A4, A5; policy engine; review queue page; daily digest
8. Sheets sync; Zapier path behind flag
9. Synthetic pilot: 10 to 20 fake investors, test mailboxes; promote to live L0
10. S1, S2, S3 SellerSequence; web form with photo upload; S4 intake
11. A7 with all six verification passes; S5; D1, D2
12. DocuSign templates; S6; deadline alerts
13. Agent eval harness (golden test sets per agent); learning loop on outcomes
14. Seller synthetic pilot; promote to live L0

Pre-launch checklist and open decisions

These are yours to do or decide; Claude Code will build around them and stop at each as a human-action checkpoint.

Pre-launch (start week 1; gates week 4 and week 8)

- [ ] Form LLC, get EIN
- [ ] Business physical address (CAN-SPAM, 10DLC)
- [ ] Business bank account
- [ ] Buy dedicated sending domain(s); set SPF/DKIM/DMARC; begin warmup
- [ ] Microsoft 365 business mailbox on the sending domain; confirm account type
- [ ] Google account and Sheet for the operator view
- [ ] Temporal Cloud account
- [ ] Railway account
- [ ] Model API key(s)
- [ ] Select and contract: deed/transaction data provider
- [ ] Select and contract: skip-trace provider (before seller phase)
- [ ] Select and contract: comps/AVM provider (before seller phase)
- [ ] Twilio account; 10DLC brand and campaign registration (before any SMS)
- [ ] DocuSign account; PSA and assignment templates reviewed by a TX attorney
- [ ] Title company per Active market that handles assignments
- [ ] Approve sender identity, outreach templates, send windows, daily caps

- [ ] Approve first Active markets from M1's initial run

Open decisions

Decision	Default if you do not decide
Assignment fee target per deal	$10,000
`market_pct` for MAO	0.70
Negotiation room	7%
Investor follow-up interval	4 days
Seller follow-up intervals	3 / 7 / 14 days
Daily send caps at L1	50 investor, 200 seller email
Soft budget cap	$1,500/mo, alert at 80%
Markets to seed M1 with	Houston plus the 20 largest Sunbelt and Midwest metros

Prompt for Claude Code

Export this doc as Markdown into the project folder alongside the two v1 Word documents, open Claude Code there, and paste the block below.

```
You are the lead engineer for a production real estate wholesaling automation
system. Three documents are in this folder:
Wholesale_Automation_Master_Spec_v1.docx (product intent),
Claude_Code_Wholesaling_Build_Guide_v1.docx (original technical plan), and
Wholesaling_Automation_Architecture_v2.md (current architecture; it
supersedes v1 wherever they conflict). Read all three completely before
writing any code.

First deliverable: REQUIREMENTS.md listing every requirement from the three
documents with a status of CONFIRMED, PROPOSED, REJECTED, or OPEN and the
source document. Never silently replace a user decision; where v2 overrides
v1, mark the v1 item REJECTED with the reason. Stop and show me this file
before continuing.

Then implement the v2 Build order in sequence. Stack: Python 3.12, FastAPI,
PostgreSQL, Temporal (Python SDK), Railway, Microsoft Graph, Google Sheets
API, provider-agnostic model layer defaulting to Anthropic. Use a multi-agent
development workflow if available with separate owners for architecture,
integrations, agent logic, QA/testing, and deployment, and integrate and
review all contributions.

Rules:
- Do not assume any credentials, accounts, or paid services exist. Create
.env.example, SETUP.md, and a human-action checkpoint for each; stop at each
checkpoint.
- Build every integration behind an interface with a mock implementation so
tests and the synthetic pilot run without real services.
- Implement the autonomy policy table exactly as in v2: same code path at
every level, policy decides send vs. queue_for_human. Default every action to
L0.
- Implement A7 with all six verification passes and the gate list as written;
the verification record is persisted with every offer.
- Every outbound send is idempotent and logged. Every agent run is logged
with prompt hash, model, tokens, cost, and output.
- No live messages, no purchases, no SMS, no contract or earnest-money
automation at any point in this build.
- Zapier parser path exists behind ZAPIER_PARSER_ENABLED, default false.
- After each milestone, report: files implemented, tests passing, blocked
items, and exact next steps. Do not claim 24/7 operation until the Railway
```

```
deployment is verified with a restart test.

Start with Build order steps 1 through 3. Ask before making any architectural
choice not covered by v2.
```